GG-G-91c
April 11, 1967
SUPERSEDING
Fed. Spec. GG-G-91b
May 23, 1960

FEDERAL SPECIFICATION

GAGE, TIRE PRESSURE, SELF-CONTAINED AND INFLATOR-GAGE, PNEUMATIC TIRE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers direct-reading tire pressure and inflator gages, calibrated in pounds per square inch.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Tire pressure and inflator gages covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - Precision gage.
Type II - General service gages.
  Class A - Heavy duty.
    Style 1 - Short extension, single-chuck foot.
    Style 2 - Long extension, dual-chuck foot.
  Class B - Light duty, single-chuck foot.
    Style 1 - Low pressure.
    Style 2 - High pressure.
    Style 3 - Low pressure, air-liquid.
Type III - Inflator-gage, internal indicating.
    Style 1 - Single-chuck foot.
    Style 2 - Dual-chuck foot.
    Style 3 - Dual-chuck foot, standard and large bore.
Type IV - Inflator gage, external indicating.
    Style 1 - Single-chuck foot
    Style 2 - Dual-chuck foot.
    Style 3 - Dual-chuck foot, standard and large bore.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:
  QQ-P-416 - Plating, Cadmium (Electrodeposited).
  PPP-B-566 - Boxes, Folding, Paperboard.
  PPP-B-636 - Box, Fiberboard.
  PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
  PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
  PPP-B-676 - Boxes, Setup.

Federal Standard:
  Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:
MIL-P-116 - Preservation, Methods of.
MIL-C-4109 - Coupling Assembly: Low Pressure, Air-Hose, Quick Disconnect.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MS-33586 - Metals, Definition of Dissimilar.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:
H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), the supplier shall furnish one complete gage for preproduction inspection. The sample may be either a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a preproduction sample shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Materials. Materials, not specifically designated, shall be compatible with the intended use and requirements specified herein. Synthetic materials shall not burn at a rate exceeding 2 inches per minute following ignition by a Bunsen burner. Materials having a cellulose nitrate base are not acceptable. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS 33586.

3.2.1 Castings and forgings. All castings and forgings shall be free from defects, scale and mismatching. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein.

3.3 Construction. The overall construction of the gages shall be free from any characteristics or defects that might render the gages unsuitable or inefficient for the intended purpose. Types I and II gages shall withstand a 25 percent, plus or minus 5 percent, overload air pressure, without damage, for a minimum of 10 minutes. Types IV and V gages shall withstand an air pressure of 175 p.s.i. for a minimum of 10 minutes. All parts subject to maintenance disassembly shall be accessible for adjustment, repair, and cleaning without the necessity of special tools. Type II, class A gages shall withstand a force of 125 pounds applied at the center of the gage without permanent deformation and shall withstand penetration of a 1/16 inch radius test pin when a force of 125 pounds is applied at any point on the barrel or stem. Slight deformation or marking not affecting serviceability or appearance is acceptable.

3.3.1 Endurance. The gages shall be so constructed as to withstand not less than 2,000 actual gaging operations for types I and II gages, and 10,000 actual gaging operations for types III and IV inflator gages at their maximum operating pressure range, without the necessity of recalibration, adjustment, or repair.

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3.3.2 Safety devices. Each tire gage shall be equipped with safety devices where practical. Parts which are hazardous to the operator shall be guarded.

3.3.3 Shock. The tire gages covered by this specification shall withstand being dropped free fall, from a minimum height of 3 feet to a concrete surface five times for types I and II gages and fifty times for types III and IV inflator gages.

3.3.4 Welding, brazing, and soldering. Welding, brazing, and soldering shall be employed only where these operations are required in the original design. Such processes shall not be employed as repair measures for defective parts.

3.3.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.6 Threads. All pipe threads shall conform to the applicable requirements of Handbook H28. All other threads shall be in accordance with the tire gage manufacturer's commercial practice.

3.3.7 Interchangeability. All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced without requiring modification. Where practicable, all such parts shall be permanently and legibly marked with the manufacturer's part number in accordance with MIL-STD-130.

3.3.8 Metal finish. All functioning surfaces of the gages shall be smooth and free from tool marks and extraneous materials. All exposed sharp edges shall be broken with a radius or chamfer, as applicable.

3.3.9 Protective finish. Unless otherwise specified (see 6.2), all metal parts shall be plated, treated, or be of corrosion resistant material to withstand the corrosive action of salt, fog, and high relative humidity as encountered in general service. Recalibration devices shall be of rust-resisting materials, but not necessarily plated. Any cadmium plating shall be accomplished in accordance with the requirements of QQ-P-416.

3.4 Design. Tire pressure gages furnished under this specification shall be convenient to use, and entirely adequate in every respect for the intended purpose. Specific design characteristics shall be as specified herein for each type, class, and style.

3.4.1 Illustrations. The illustrations shown on figures 1 through 8 herein are for the convenience of identification and are not intended to preclude the purchase of gages which are otherwise in accordance with the requirements of this specification.

3.4.2 Chuck foot. Unless otherwise specified (see 6.2), the chuck foot of all gages shall be offset and at an angle between 15 degrees and 35 degrees to the centerline of the gage barrel. The chucks shall be so constructed as to fit standard (0.302 - 32) threads used on tire valves, or when hereinafter defined as large bore, the chuck foot shall fit (0.482 - 26) valve threads used on off-road tires. When applied to the valve stem, either chuck shall depress the valve core plunger permitting the free passage of air from the tire into the gage pressure chamber, without loss of air or damage to the stem or core members. The chuck foot shall be made integral with the extension or connecting member, or rigidly attached thereto in such a manner as not to become loose or detached.

3.4.2.1 Single-chuck foot. This type foot shall be provided with a single chuck which shall be so constructed as to operate the gage when depressed on the valve stem. The chuck for type II gages shall also be provided with an external deflator pin.

3.4.2.2 Dual-chuck foot. This type of foot shall be provided with a dual opposed deflator pin. The dual-chuck foot assembly shall be so constructed as to operate the gage from one chuck and render the other chuck deflator pin inoperative. The deflator pin shall be hardened steel and shall be constructed so that distortion of the rubber sealing washer cannot partially or completely block the flow of air through the gage.

3.4.3 Types I and II gages. Gages of types I and II shall be composed primarily of a barrel with indicator bar and single or dual chuck foot with extension, except that type II, class B, shall have no extension.

3.4.3.1 Barrel. The barrel shall be so constructed as to fully enclose and shield the internal working mechanism. The inside of the barrel shall be treated to prevent corrosion along its entire length.

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3.4.3.2 Indicator bar. The indicator bar shall be furnished in round, rectangular, triangular, or square shapes. The connection between the indicator bar and the internal actuating mechanism shall be such as to permit the indicator bar to remain at the indicated pressure after the source of pressure has been removed. The graduation marks shall be stamped, etched, or otherwise set to a width and depth to provide a permanency at least equal to the life of the gage. The accuracy in graduation spacings shall be such as to meet with the requirements of each particular type and class. The numerals indicating the applicable graduation increment shall be a minimum 1/16 inch in height. Both graduations and numerals shall be clear and well defined to facilitate reading of the gage under normal service conditions.

3.4.3.3 Type I precision gage. Type I precision gage shall have a low pressure reading of 10 pounds per square inch gage (p.s.i.g.) or less, and a high pressure reading of not less than 160 p.s.i.g. The indicator bar shall be graduated in 2 p.s.i. increments, or less, the entire length of the scale. The scale shall be marked with the appropriate numerical value at each 10 p.s.i. interval. The graduated length of the scale shall be not less than 2-1/2 inches nor more than 3-1/2 inches. The gage shall be 12 inches, plus or minus 2 inches, in overall length and shall be provided with an extension 3-1/2 to 4-1/2 inches in length having a standard bore dual-foot chuck. Each gage shall be provided with positive identification to avoid mistaking the precision gage and general purpose gage. Gages of this type shall be similar in design to figure 1.



Figure 1. - Type I precision gage

3.4.3.3.1 Accuracy. Precision gages, at plus 70°F. and minus 20°F., and within 5 seconds after pressure has been applied, shall indicate the true pressure within the following limits:

Accuracy at plus 70°F., plus or minus 5°F. (room temperature), to be within plus or minus:

1 p.s.i. at 30 p.s.i.g.
1 p.s.i. at 50 p.s.i.g.
1.5 p.s.i. at 75 p.s.i.g.
2 p.s.i. at 120 p.s.i.g.

Accuracy at minus 20°F. to minus 25°F. to be within plus or minus:

1 p.s.i. at 30 p.s.i.g.
1.5 p.s.i. at 50 p.s.i.g.
2 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

3.4.3.4 Type II, general service gage. Type II general service gages shall be of two classes for general service usage, as specified herein.

3.4.3.4.1 Type II, class A heavy-duty general service gage. Gages of this type and class shall be of heavy construction throughout to withstand rough usage as encountered in service stations and tire shops. The type II, class A gages shall have a low pressure reading of 20 p.s.i.g. or less, and a high pressure reading of not less than 130 p.s.i.g. The indicator bar shall be graduated in 2 p.s.i. increments or less up to at least 60 p.s.i.g. on the scale and in 5 p.s.i.g. increments or less for the remainder of the scale. The scale shall be marked with the appropriate numerical value at each ten pound unit or less. The length of the graduated scale shall not be less than 3-1/4 inches.

3.4.3.4.1.1 Accuracy. Gages of this type shall indicate the true pressure at plus 70°F., and minus 20°F. and within 5 seconds after pressure has been applied, within the following limits:

Accuracy at plus 70°F. plus or minus 5°F. (room temperature) to be within plus or minus:

1.5 p.s.i. at 30 p.s.i.g.
1.5 p.s.i. at 50 p.s.i.g.
2 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

Accuracy at minus 20°F. to minus 25°F. to be within plus or minus:

1.5 p.s.i. at 30 p.s.i.g.
2 p.s.i. at 50 p.s.i.g.
3 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

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3.4.3.4.1.2 Type II, class A, style 1, short-extension single-chuck foot gage. Gages of this type, class, and style shall be 8-1/2 to 9-1/2 inches in overall length, and provided with an extension 1-1/2 to 2-1/2 inches in length having a standard bore single-chuck foot. These gages shall be similar in design to figure 2.

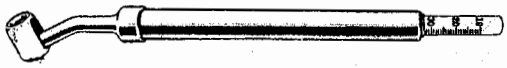


Figure 2. - Type II, class A, style 1 general service gage.

3.4.3.4.1.3 Type II, class A, style 2, long-extension dual-chuck foot gage. Unless otherwise specified (see 6.2), the gages of this type, class, and style shall be not less than 11 inches in overall length. Each gage shall have an extension of not less than 5-1/2 inches in length, a standard bore dual-chuck foot, and provisions for hanging up the gage. These gages shall be similar in design to figure 3.

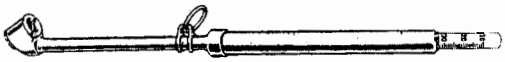


Figure 3. - Type II, class A, style 2 general service gage.

3.4.3.4.2 Type II, class B, light-duty gage. Gages of this type and class shall be of three styles, and shall be for light-duty, individual usage. The type II, class B gages shall have a pocket clip, a standard bore single-chuck foot, and shall be 5 to 6 inches in overall length.

3.4.3.4.2.1 Accuracy. Type II, class B, styles 1 and 3 gages shall indicate the true pressure within plus 0 p.s.i. or minus 2 p.s.i., at plus 70°F. plus or minus 5°F. temperature at 40 p.s.i.g. pressure. Type II, class B, style 2 gages shall indicate the true pressure at the same temperature within plus 1 p.s.i. or minus 2 p.s.i. at 75 p.s.i.g. reading, and within plus or minus 3 p.s.i. at 100 p.s.i.g. reading. All type II, class B gages shall indicate the true pressure at minus 20°F. to minus 25°F. temperature within plus or minus 3 p.s.i. at 40 p.s.i.g. pressure. Gages shall indicate the true pressure at either of the above temperatures within the aforementioned limits within 5 seconds after the pressure has been applied.

3.4.3.4.2.2 Type II, class B, style 1, low-pressure gage. Gages of this type, class, and style shall have a low pressure reading of 10 p.s.i.g. or less, and a high pressure reading of not less than 50 p.s.i.g. The indicator bar shall be graduated in 1 p.s.i. increments, with the appropriate numerical value at each five pound unit or less. These gages shall be similar in design to figure 4A.



Figure 4A. - Type II, class B, style 1 general service gage.

3.4.3.4.2.3 Type II, class B, style 2, high-pressure gage. Gages of this type, class and style, shall have a low pressure reading of 20 p.s.i.g. or less, and a high pressure reading of not less than 100 p.s.i.g. The indicator bar shall be graduated in 5 p.s.i. increments maximum, with the appropriate numerical value at each 10 p.s.i. interval. These gages shall be similar in design to figure 4B.

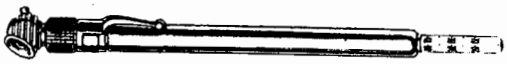


Figure 4B. - Type II, class B, style 2 general service gage.

3.4.3.4.2.4 Type II, class B, style 3, air and liquid gage. Gages of this type, class, and style shall have a minimum pressure range of 12 p.s.i.g. to 45 p.s.i.g. The indicator bar shall be graduated in 1 p.s.i. increments, with the appropriate numerical value at each five pound unit or less. The gage shall be fully protected against mud, sand, dust, brine, antifreeze solutions and other harmful materials. The indicator bar shall be attached to the internal mechanism in such a manner as to facilitate flushing and oiling of the internal mechanism by pump action of the indicator bar. A hole shall be provided in the outer end of the barrel to permit injection and ejection of a liquid cleaner and light oil. These gages shall be similar in design to figure 4C.

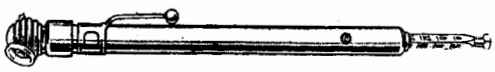


Figure 4C. - Type II, class B, style 3 general service gage.

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3.4.4 Types III and IV inflator gages. Types III and IV inflator gages shall consist essentially of a pneumatic tire pressure gage, a control valve and an attached length of hose equipped for supplying compressed air to a valve stem by means of a trigger or lever mechanism. The gage shall indicate the air pressure upon release of the trigger or lever. The trigger or lever shall be located in a manner to facilitate the operation of the inflation attachment when the operator is wearing heavy gloves or mittens. The trigger or lever shall be hinged at, or near, one end and shall develop a leverage ratio of not less than 3 to 1 on the air-valve assembly. All types III and IV gages shall be constructed to withstand a pressure of 175 p.s.i.g. without leakage. The gage shall have a low pressure reading of 10 p.s.i.g. or less and a high pressure reading of not less than 120 p.s.i.g., with graduations each 2 p.s.i., or less, the entire length of the scale. Each 10 p.s.i. increment shall be clearly marked with the appropriate numerical value.

3.4.4.1 Accuracy. Accuracy requirements for types III and IV gages shall be as follows: Construction shall be such as to insure accuracy within the tolerance of plus or minus 1/2 of 1 percent of the full-scale reading at the 30 p.s.i. calibration, and within the tolerance of plus or minus 2 percent of the full-scale reading at the 80 p.s.i. calibration when subject to temperatures of from plus 110°F. to plus 115°F. At temperatures of minus 20°F., the accuracy shall be within the tolerance of plus or minus 1 percent of the full-scale reading at the 30 p.s.i. calibration, and within the tolerance of plus or minus 3 percent of the full-scale reading at the 80 p.s.i. calibration.

3.4.4.2 Discharge hose. Gages of types III and IV shall be provided with a 1/4 inch inside diameter (I.D.) discharge hose not less than 14 inches in length and in accordance with tire gage manufacturer's commercial practice. The hose shall be fitted on one end with a male threaded connector and on the other end with a full swivel connector.

3.4.4.3 Extension and foot. Gages of types III and IV, styles 2 and 3 shall be furnished with a metal extension not less than 5-1/2, nor more than 6-1/2 inches, in overall length. The extension shall be fitted on one end with a female threaded connector and on the other end with a dual-chuck foot of the size as hereinafter defined for each type, class, and style gage.

3.4.4.4 Couplings and connectors, quick-disconnect. Gages of types III and IV, style 3 shall be furnished with quick-disconnect couplings and connectors in accordance with MIL-C-4109. The couplings shall be class A, style 2. The connectors shall be class A, style 1.

3.4.4.5 Type III, internal indicating inflator gage. Gages of this type shall be of three styles, and shall be internal indicating with maximum graduations of 2 p.s.i. increments. The internal incicating and air control assemblies shall be separate self-contained units, removable and replaceable without the use of special tools. Unless otherwise specified (see 6.2), the air valve element, gage cartridge, and sealing washers shall comply with figure 5.

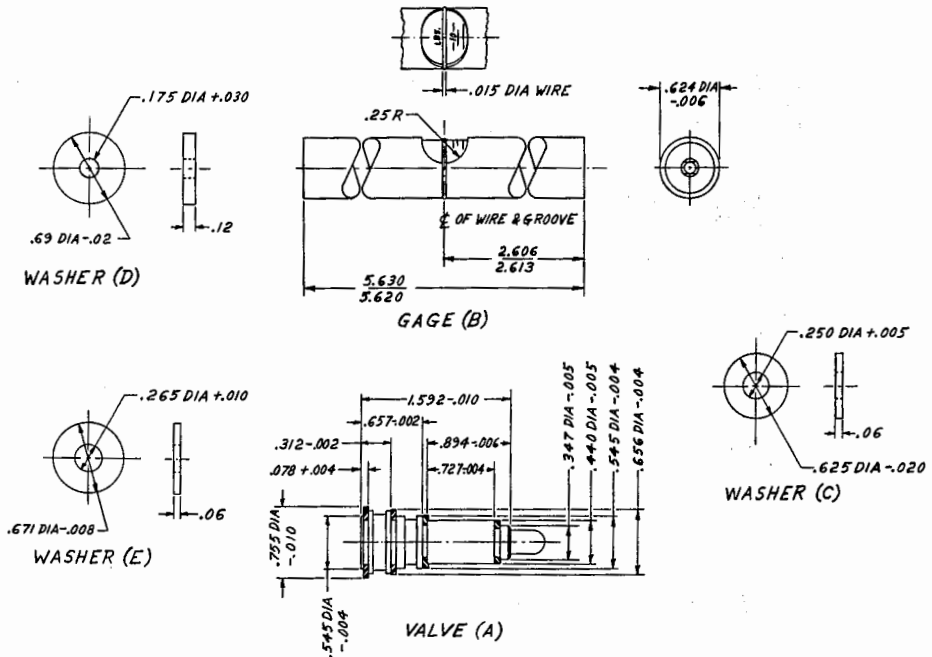


Figure 5. - Gage cartridge, valve element and sealing washers for type III gages.

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3.4.4.5.1 Body. The body of type III gages shall be of cast aluminum alloy or similar metal and furnished with an air inlet, outlet, air screen, and valve actuating lever or trigger. The air inlet opening shall be threaded to receive a 1/4 inch NPT male fitting. A wire-cloth air screen, 40 mesh (.010 inch), shall be furnished for the inlet opening and readily accessible for cleaning. The valve actuating lever shall be hinged at, or near, one end, and shall develop a leverage ratio of not less than 3 to 1 on the air valve assembly.

3.4.4.5.2 Type III, style 1, single-chuck foot gage. Gages of this type and style shall be furnished with a standard bore single-chuck foot.

3.4.4.5.3 Type III, style 2, dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and shall be similar in design to figure 6.

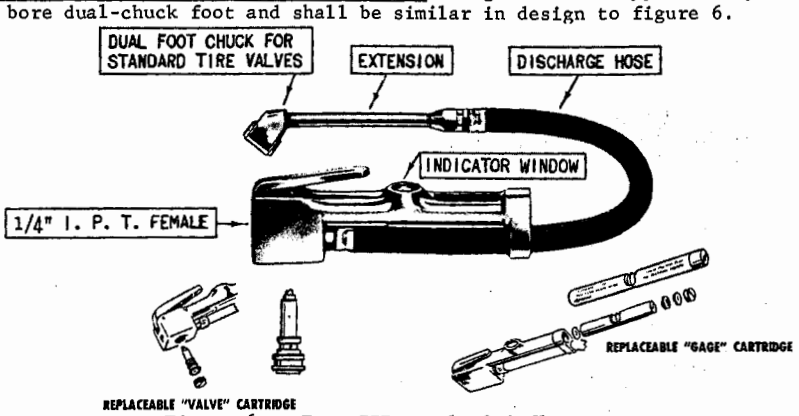


Figure 6. - Type III, style 2 inflator gage.

3.4.4.5.4 Type III, style 3, standard and large bore dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and a large bore dual-chuck foot. Both the dual footings shall be removable and replaceable at the end of the discharge hose by means of quick-disconnect fittings (see 3.4.4.4). Each of the dual footings shall be attached by a chain with rings to prevent the loss of either chuck while not in use. The ring shall be so constructed as to slide freely over the discharge hose, and shall be of such diameter as not to pass over the coupler. These gages shall be similar in design to figure 6A.

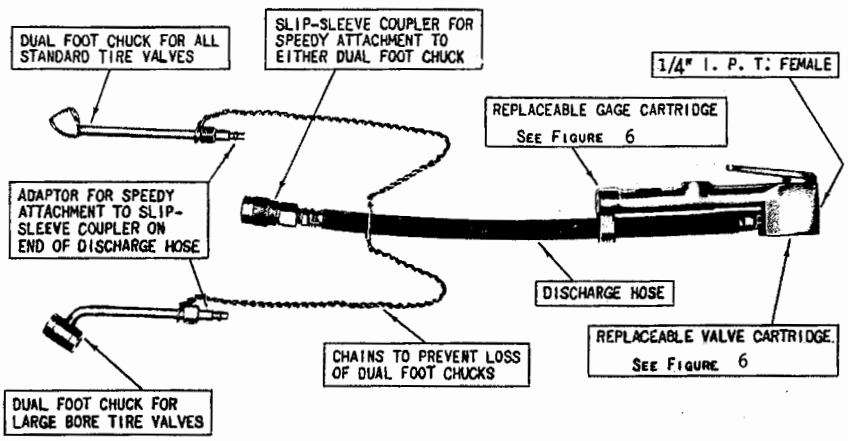


Figure 6A. - Type III, style 3 inflator gage.

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3.4.4.6 Type IV, external indicating inflator gage. Gages of this type shall be of three styles and shall be external indicating with maximum graduations of 2 p.s.i. the entire length of the scale.

3.4.4.6.1 Body. The body of type IV gages shall be of cast aluminum alloy or similar metal and furnished with an air inlet with screen, outlet and valve control lever or trigger. The air inlet opening shall be threaded to receive a 1/4 inch NPT male fitting. A wire-cloth air screen, 40 mesh (0.010 inch), shall be furnished for the inlet opening and readily accessible for cleaning. The air outlet opening shall be threaded to receive the NPT male fitting of the discharge hose.

3.4.4.6.2 Type IV, style 1, single-chuck foot gage. Gages of this type and style shall be furnished with a standard bore single-chuck foot.

3.4.4.6.3 Type IV, style 2, dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and be similar in design to figure 7.

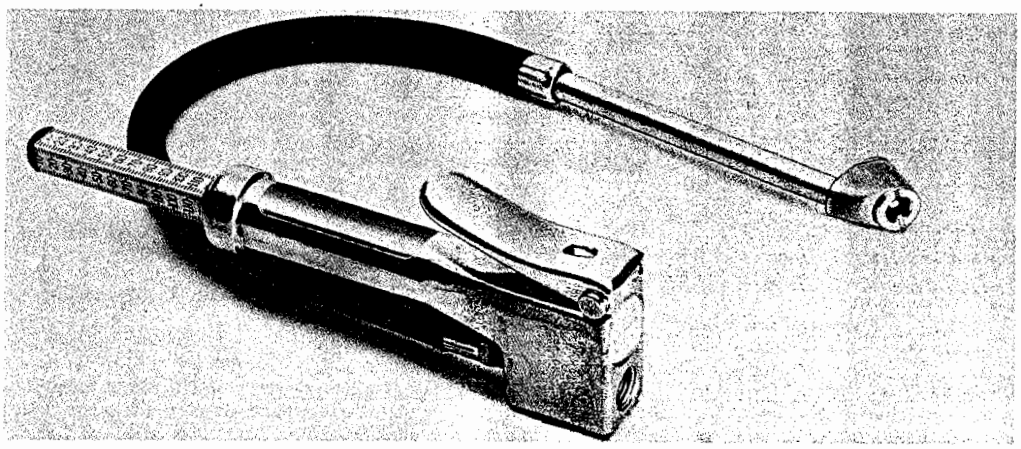


Figure 7. - Type IV, style 2 inflator gage.

3.5.5.6.4 Type IV, style 3, standard and large bore dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and a large bore dual-chuck foot. Both dual footings shall be removable and replaceable at the end of the discharge hose by means of quick-disconnect fittings (see 3.4.4.4). Each of the dual footings shall be attached by a chain with rings to prevent the loss of either chuck while not in use. The ring shall be so constructed as to slide freely over the discharge hose, and shall be of such diameter as not to pass over the coupler. These gages shall be similar in design to figure 8.

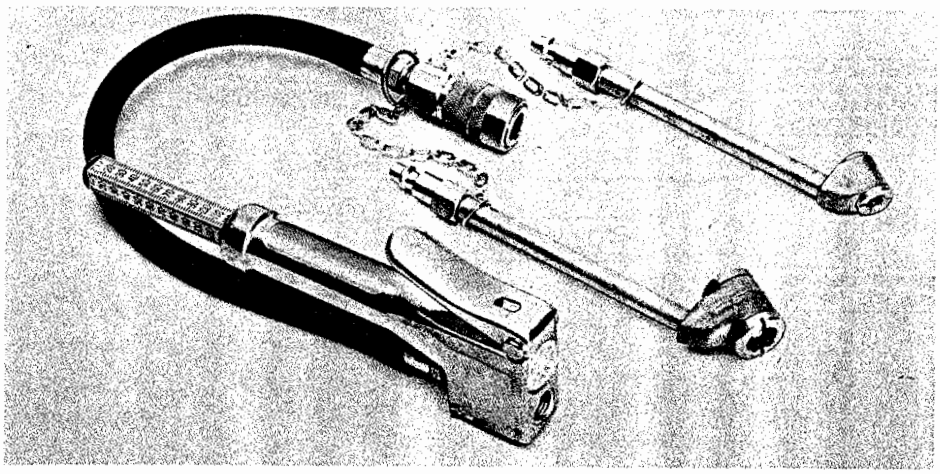


Figure 8. - Type IV. - Style 3 inflator gage.

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3.5 Storage temperatures. All gages and all components thereof shall be able to withstand without damage, storage in a temperature of minus 65°F. for periods up to 72 hours, and a 24-hour cycle of temperature, starting at 10 hours at plus 90°F., followed by 5 hours of steady decrease to plus 155°F., 4 hours at plus 155°F., and 5 hours of steady decrease to plus 90°F. Following these exposures, and after being stabilized at a temperature of plus 70°F. plus or minus 5°F., any gage of any type, class, or style shall be capable of meeting the applicable accuracy and performance requirements set forth in this specification.

3.6 Identification of product. Each tire gage, except as noted below, shall be permanently and legibly marked to give the following desired information: Manufacturer's name or trademark and the Federal Stock Number. In the case of type II, class B gages, there shall only appear the manufacturer's name or known trademark. In the case of types III and IV gages, there shall appear a serial number, as well as the manufacturer's trademark and the Federal Stock Number. When specified (see 6.2), special markings of the gages shall be required. On the box or container for each gage, there shall appear the following information:

Gage tire pressure ----------.
Type ----------.
Class ----------.
Style ----------.
Fed. Spec. No. GG-G-91c ----------.
Mfr. Model No. ----------.
Order No. ----------.
Stock No. ----------.
Manufacturer's name or trademark U.S. Property ----.

3.7 Instruction leaflet. Each type II, class B, style 3, and Type III gage shall be furnished with a leaflet covering care and operation. The leaflet shall be that normally furnished by the contractor in general commercial practice.

3.8 Workmanship. Workmanship shall be first class and of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements in section 3. All components, subassemblies, parts, and final assembly shall comply with tests in section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and tests. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection.
(b) Quality conformance inspection.
(c) Inspection of preparation for delivery.

4.2.1 Preproduction inspection. Preproduction inspection applies to the preproduction sample noted in 3.1. Unless otherwise specified (see 6.2), preproduction inspection shall consist of the examination in 4.3 and all the tests under 4.4. Failure of the preproduction sample to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to each gage prior to being submitted for acceptance under the contract. The inspection shall consist of the examination in 4.3, accuracy tests in 4.4.1, and inspection of preparation for delivery in 4.5. Failure of the unit to pass examination or test shall be cause for rejection.

4.2.2.1 Sampling. When specified (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in the contract. Samples for the test shall be selected at random from the sample that passed the examination in 4.3.

4.3 Examination. Each production item, or sample, shall be examined to determine compliance with the requirements in section 3, using the following classification of defects:

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Classification of defects

Major defects
(AQL shall be 2.5 percent defective)

1. Defective casting or forging affecting operation (see 3.2.1).
2. Protective finish not as specified (see 3.3.9).
3. Chuck foot not as specified (see 3.4.2).
4. Dual-chuck foot not as specified (see 3.4.2.2).
5. Indicator bar not as specified (see 3.4.3.2).
6. Pressure range not as specified (see 3.4.3.3, 3.4.3.4.1, 3.4.3.4.2.2, 3.4.3.4.2.3, 3.4.3.4.2.4, and 3.4.4).
7. Indicator bar markings not as specified (see 3.4.3.2, 3.4.3.3, 3.4.3.4.1, 3.4.3.4.2.2, 3.4.3.4.2.3, 3.4.3.4.2.4, 3.4.4, 3.4.4.5, and 3.4.4.6.
8. Discharge hose not as specified (see 3.4.4.2 and 3.4.4.4).
9. Air screen not as specified (see 3.4.4.5.1).

Minor defects
(AQL shall be 4.0 percent defective)

51. Fastening devices not as specified (see 3.3.5).
52. Gages with tool marks or rough finish (see 3.3.8).
53. Identifying sleeve not as specified (see 3.4.3.3).
54. Dimensions not as specified (see 3.4.3.3, 3.4.3.4.1.2, 3.4.3.4.1.3, and 3.4.3.4.2).
55. Valve control lever or trigger inoperative (see 3.4.4).
56. Identification marking illegible (see 3.6).
57. Instruction leaflet not as specified (see 3.7).

4.4 Tests.

4.4.1 Accuracy test. Gages shall be tested at a temperature of plus 70°F. plus or minus 5°F., and shall indicate the true pressure within the limits of accuracy required of the particular type, class, and style of gage being tested in relation to the temperature referenced above. An air pressure measuring device having an overall accuracy of one-fourth of 1 percent of full-scale value shall be used as a criterion (see 3.4.3.3.1, 3.4.3.4.1.1, 3.4.3.4.2.1, and 3.4.4.1).

4.4.2 Construction test.

4.4.2.1 Overload test. Types I and II gages shall be subjected to an overload of 25 percent plus or minus 5 percent in excess of its rated capacity for a minimum of 10 minutes. Types III and IV inflator gages shall be subjected to a pressure of 175 p.s.i.g. for a minimum period of 10 minutes. The gage shall then be subjected to the applicable accuracy test as outlined in 4.4.1 (see 3.3).

4.4.2.2 Concentrated load test. Type II, class A gages shall be subjected to a concentrated load test. Suspend the gage on 5/8 inch diameter test pins set 7 inches apart for the style 1 gage and 9 inches apart for the style 2 gage, suspend another 5/8 inch diameter pin on the gage at the center and slowly apply a force up to 125 pounds. The gage shall then be examined for damage. This test shall not effect accuracy as outlined in 4.4.1 (see 3.3).

4.4.2.3 Indentation test. Type II, class A gages shall be subjected to an indentation test. Place the gage barrel or stem on a flat surface and slowly apply a load up to 125 pounds through a 1/16 inch radius test pin or mandrel. The gage barrel or stem shall then be examined for deformation. This test shall not effect accuracy as outlined in 4.4.1 (see 3.3).

4.4.3 High and low temperature operational test. Gages shall be tested at the temperatures and air pressures specified in the applicable requirement for accuracy and shall indicate the true pressure within the limits of accuracy required for the particular type, class, and style of gage being tested. An air pressure measuring device having an overall accuracy of one-fourth of 1 percent of full-scale value shall be used as a criterion. All types I and II gages shall indicate the true pressure within the specified tolerances at the applicable temperatures and pressure within 5 seconds after pressure has been applied (see 3.4.3.3.1, 3.4.3.4.1.1, 3.4.3.4.2.1, and 3.4.4.1).

4.4.4 Storage test.

4.4.4.1 Cold test. The gage shall be placed in an ambient air temperature of minus 60°F. to minus 65°F. for a period of 72 hours. At the end of this period, the gage shall be allowed to stabilize at an air temperature of 70°F. plus or minus 5°F. The gage shall then be inspected for evidence of damage and tested in accordance with 4.4.1 (see 3.5).

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4.4.4.2 Heat test. The gage shall be subjected to a 24-hour cycle of temperature beginning with 10 hours at plus 90°F. plus or minus 5°F., followed by 5 hours of steady increase to plus 150°F. to plus 155°F., 4 hours at plus 150°F. to plus 155°F., and 5 hours of steady decrease to plus 70°F. plus or minus 5°F. The gage shall then be inspected for evidence of damage and tested in accordance with 4.4.1 (see 3.5).

4.4.5 Endurance test. The types I and II gages shall be actuated by the application of air pressure and the indicator bar closed after each application of air, for not less than 2,000 gaging operations or cycles. The types III and IV gage shall be actuated by the application of air pressure and the air valve operated so as to return the gage to zero after each application of air, for not less than 10,000 gaging operations or cycles. The type I and type II, class A gages shall be operated from an air source of 150 to 160 p.s.i.g. The type II, class B, styles 1 and 3 gages shall be operated from an air source of 45 to 50 p.s.i.g. The type II, class B, style 2 gage shall be operated from an air source of 90 to 100 p.s.i.g. The types III and IV gages shall be operated from an air source of 150 p.s.i.g. and the lever or trigger shall be actuated not less than 10,000 times to assimilate the required number of gaging operations. The gaging operations or cycles shall be at a rate of approximately 10 per minute for all types of gages. The method of applying air to the gages, actuating the indicator bar on the types I and II gages, and actuating the lever or trigger on the types III and IV gages is at the discretion of the testing facility. The gage shall then be examined for damage and the accuracy shall be tested as outlined in 4.4.1 (see 3.3.1).

4.4.6 Shock test. The tire gage shall be dropped from a vertical distance of 3 feet onto a concrete surface five times for types I and II gages and 50 times for types III and IV gages. The gage shall be examined for damage, and the accuracy shall be tested in accordance with 4.4.1 to determine compliance with 3.3.3. Failure of any gage to pass this test shall be cause for rejection.

4.4.7 Flammability test. Synthetic materials for test shall be extended outward and clamped 1/4 inch above a 5 inch square piece of 10-mesh Bunsen burner gauze in a horizontal position. A Bunsen burner with an efficient flame 1/2 to 3/4 inch in height, shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the sample. At the end of 30 seconds, the flame shall be removed and the material allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the sample after the removal of the Bunsen burner. In case the material does not continue to burn after the first ignition, the burner shall be replaced under the free end for a period of 30 seconds immediately following the extinction of the flame. If the sample does not continue to burn until the flame has reached the end of the material after the second ignition, the sample shall be considered self-extinguishing. When tested as described, the synthetic materials shall not burn at a rate exceeding 2 inches per minute (see 3.2).

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the gages shall be examined and tested to determine conformance with the requirements of section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105, and an AQL of 4.0 percent defective.

4.6 Defective units. Units found defective in the course of inspection shall be corrected to the satisfaction of the Government or shall be replaced with nondefective units.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The gages shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Preservation. Types I and II gages shall be individually preserved in accordance with method IC-1 and types III and IV gages shall be individually preserved in accordance with method IC-2 of MIL-P-116.

5.1.1.3 Intermediate packaging. Types I and II gages of identical description shall be intermediate packaged in snug fitting fiberboard boxes conforming to W5c of PPP-B-636. The boxes shall be closed in accordance with the appendix to the container specification. Types III and IV gages packaged as specified in 5.1.1.2 require no intermediate packaging.

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5.1.2 Level B.

5.1.2.1 Unit packaging. Each tire pressure gage shall be packaged in a paperboard box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.2.2 Intermediate packaging. Ten unit packages of tire pressure gages of like description shall be intermediate packaged in a paperboard or fiberboard box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636. The fiberboard box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. Gages shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The preserved and packaged gages shall be packed in fiberboard boxes conforming to class 2 of PPP-B-640. Boxes shall be closed and strapped in accordance with the appendix to the container specification. Exterior shipping container shall accommodate the greatest number of items in the least number of containers.

Note: Intermediate containers may be used as the shipping container, if the total shipment does not exceed the maximum weight limitations of the container specification.

5.2.2 Level B. Preserved and packaged gages shall be packed and marked as specified in 5.2.1.

5.2.3 Civil agencies. Five intermediate packages of tire pressure gages of like description shall be packed in a close-fitting box conforming to PPP-B-636, class - domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.4 Level C. The gages shall be packed to afford adequate protection against damage during shipment from the supply source to the first receiving activity. Shipping containers shall comply with rules and regulations of carriers as applicable to the mode of transportation.

5.3 Standard pack (civil agencies). The standard pack for civil agency procurements shall be as specified in 5.1.2 and 5.2.3 (see 6.4).

5.4 Marking.

5.4.1 Military agencies. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.4.2 Civil agencies. In addition to any special marking specified in the procurement document, shipments shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use.

6.1.1 Type I. This precision gage is intended for use to check the accuracy of general service gages.

6.1.2 Type II, class A. This general service gage is intended for use in service areas to measure or check the pressure of air in tires or similar items.

6.1.3 Type II, class B. This general service gage is intended for use as a personal item for frequent measuring or checking the pressure of air in tires or similar items. It is intended to be carried in a person's pocket or storage compartment of a vehicle. It is not intended for rough usage or handling as often found in service areas.

6.1.4 Types III and IV. These inflator gages are intended for use on air supply lines to inflate, deflate and measure or check the pressure of air in tires or similar items. They are intended to simplify the servicing of tires and similar items into one operation.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2.1).
(c) Preproduction inspection when required and sample quantity, if different (see 3.1).
(d) Dull, dark, and nonreflective finish, when required (see 3.3.9).
(e) Special plating or treatment, when required (see 3.3.9).

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(f) Straight foot, when specified (see 3.4.2).
(g) Gage overall length, if different (see 3.4.3.4.1.3).
(h) Internal mechanism, if different (see 3.4.4.5).
(i) Special marking of the gages, if required (see 3.6).
(j) Preproduction inspection, if different (see 4.2.1).
(k) Quality conformance inspection, if different (see 4.2.2).
(l) Sampling for lot acceptance, when required (see 4.2.2.1).
(m) Selection of applicable level of preservation, packaging and packing required (see 5.1 and 5.2).

6.3 Classification change. Type III, classes A and B gages have been deleted from this specification. Type IV has been added.

6.4 Standard pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurements of Stores Stock replenishments. Procuring offices should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

Custodians:
Army - WC
Navy - YD
Air Force - 82

Review activities:
Army - MO, WC
Navy - YD
Air Force - 82

User activities:
Army - WC
Navy - MC, SH, WP

Preparing activity
Army - WC

Civilian agencies' Interest
GSA - FSS
INT - MIN

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